EMMANUEL MATTHEW J

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PROFESSIONAL SUMMARY

Highly motivated and quick-learning AI & Machine Learning Engineer skilled in Python, Machine Learning, and Deep Learning. Dedicated to creating intelligent systems through data-centric methodologies and solving practical issues through AI. Excellent analytical and logical reasoning skills along with software development fundamentals. Highly adaptable quick-learner with great communication skills and teamwork capabilities.

EDUCATION

- Petit Seminaire Higher Secondary School, Puducherry — 2022-23
  12th, Computer Science
  Percentage: 91.5
- Sri Manakula Vinayagar Engineering College, Puducherry — 2023-27
  B.Tech. Computer Science and Engineering
  Current GPA: 9.00

SKILLS

- **Programming Languages:** Python, SQL
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation
- **Libraries & Frameworks:** Scikit-learn, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
- **Web Technologies:** HTML, CSS, React, Flask, Django
- **Databases:** MySQL, PostgreSQL, Supabase
- **Tools & Platforms:** Git, GitHub, VS Code Google Colab

TECHNICAL PROFILES

- https://github.com/ematty246
- linkedin.com/in/emmanuel-matthew-j-247186328/

CERTIFICATES

- **NPTEL** – Python for Data Science | Score: 60% — **July-Aug 2024**
- **NPTEL** – Programming in Java | Score: 80% — **July-Oct 2025**

PROJECTS

(I) AI Career Recommendation System for Students

• Developed an AI-based system that analyses student resumes and recommends suitable career paths.
• Built the frontend interface and backend services to process resume data and generate recommendations.
• Integrated a Large Language Model (Meta) through API calls to provide career insights.

(II) Driver Drowsiness Monitoring System

- Developed a real-time driver monitoring system using YOLOv8 and OpenCV to detect drowsiness, yawning, mobile usage, and drinking behaviour.
- Built a Python-based desktop application with live video processing, confidence smoothing, and alert triggering logic.
- Implemented a Flask REST API to stream detection results and integrated a React Native mobile app for live status tracking and location sharing.
- Designed an alert system that triggers hardware signals via serial communication (Arduino) when drowsiness is detected continuously.
- Tech Stack: Python, YOLOv8, OpenCV, Flask, React Native, Tkinter, PySerial

(III) Fleet Maintenance Prediction System

- Developed an end-to-end machine learning system to predict vehicle maintenance requirements using fleet operational data and real-time vehicle parameters.
- Trained and optimized an XGBoost classification model using 15+ features, including engine temperature, oil quality, brake wear, tire pressure, and usage hours.
- Built a Flask REST API to serve real-time maintenance predictions with probability scores through a RESTful interface.
- Designed a responsive web application using HTML, CSS, and JavaScript for user-friendly data input and instant maintenance prediction.
- Implemented the complete machine learning pipeline, including data preprocessing, feature engineering, model training, evaluation, and model deployment.
- Tech Stack: Python, XGBoost, Scikit-learn, Flask, Pandas, NumPy, HTML, CSS, JavaScript

ACHIEVEMENTS

• Awarded Top 3 position in the 1M1B sus-AI-nability Challenge 2025, sponsored by IBM SkillsBuild, recognizing innovative AI-driven solutions that significantly contribute to sustainability.
• Received a certificate of appreciation for achieving second place at the Project Expo held during MECHATRONZ FEST 2K24, an Inter-Department Technical Symposium, hosted by the Mechatronics Department of Sri Manakula Vinayagar Engineering College, on April 12–13, 2024.

ONGOING RESEARCH WORK

This current research manuscript is titled: "*An Explainable Machine Learning Framework for Predicting Sustainable Consumer Behaviour in Neobanks: A Case Study.*" Through this study I will examine the development of machine-learning models that have been designed not only to be interpretable, but also to analyze and predict how consumers will behave with respect to sustainable financial practices.